

Ruoyun Li

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EDUCATION

Yale University M.S. in Health Informatics in Public Health	08/2025-05/2027
University of Virginia (UVA) B.S. in Statistics & Psychology, Highest Distinction	08/2021-05/2025 GPA: 3.95/4.00

RESEARCH INTERESTS

High-dimensional data analysis, Electronic Health Records, Brain Connectomics and Neuroimaging, Population Health

THESIS & PUBLICATIONS

- 1 **Li, R.**, Sequeira, S. & Zhang, A. "Latent Dimensions of Social Determinants of Health and Their Associations with Youth Psychopathology Over Time." *Manuscript in preparation*.
- 2 Liu, P., **Li, R.**, Zhao, Y. "A Cost-aware Framework for Personalized Feature Acquisition in Multi-Disease Prediction." *Manuscript in preparation*.
- 3 Wang, T., Lin, J., Zhang, X., Xiao, W., Li, Y., Chen, Y., Peng, J., Xing, H., **Li, R.**, et al. "HealthResearchBench: Reusable Test Beds for Reproduction and Independent Improvement in Healthcare and Biomedical Research." *Submitted to AAAI 2027; under review*.
- 4 Chadha, S., Tsang, S., Liu, C., **Li, R.**, Henry, T., Oishi, S., & Wood, A. "Dynamics of time spent with social others in a new social environment." *Manuscript in preparation*.
- 5 **Li, R.**, "New Places, New Feelings: How Primal World Beliefs and Mobility Behaviors Predict Momentary Subjective Experiences." *Distinguished Majors Thesis in Psychology*, University of Virginia, Advisor: Prof. Adrienne Wood & Dr. Sareena Chadha

CONFERENCE PRESENTATIONS

- 1 **Li, R.**, Sequeira, S. & Zhang, A. (2026, June). "High-Dimensional Mediation of Social Determinants of Health and Youth Psychiatric Symptoms via Functional Connectivity." Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting, Bordeaux, France
- 2 **Li, R.**, Wood, A., & Chadha, S. (2025, April). "New Places, New Feelings: How Primal World Beliefs and Mobility Behaviors Predict Momentary Subjective Experience." Poster presented at the L. Starling Reid Undergraduate Psychology Research Conference, Charlottesville, VA

RESEARCH EXPERIENCE

Post-Training and Data Filtering for a Medical Foundation Model 07/2026 – Present
Supervisor: Prof. Hua Xu, Department of Biomedical Informatics & Data Science, Yale School of Medicine New Haven, CT

- Developed data filtering and post-training strategies for a medical foundation model trained on longitudinal claims data, focusing on high-quality cohort construction and instruction fine-tuning
- Designed EHR-specific information-density and data-quality filters to prioritize clinically informative patient trajectories

HealthResearchBench: Reusable Test Beds for Healthcare and Biomedical Research 05/2026 – Present
Collaborative Group Research Project

- Contributed to the experimental schema and evaluation protocol for agent-based reproduction of healthcare and biomedical research
- Guided AI agents in reproducing TransUNet medical image segmentation and the MIMIC-III multitask benchmark, auditing experimental setup, data processing, model execution, and reproduced results for scientific validity
- Analyzed agent performance and failure modes across reproduction experiments; co-authored the resulting manuscript submitted to AAAI 2027

Multilayer Hypergraph Modeling of Multiorgan Pathways and Chronic Disease Risk 07/2025 – Present
Supervisor: Prof. Yize Zhao, Department of Biostatistics, Yale School of Public Health New Haven, CT

- Developing a multilayer hypergraph framework that represents behavioral, social, genetic, organ-imaging, and chronic disease variables as distinct but interconnected layers
- Modeling within-layer associations and learning sparse, interpretable cross-layer hyperedges that capture higher-order exposure-organ-disease pathways
- Integrating causal mediation and Mendelian randomization to evaluate how upstream risk factors influence chronic disease through multiorgan pathways

Cost-Aware Framework for Personalized Feature Acquisition in Multi-Disease Prediction 07/2025 – Present
Supervisor: Prof. Yize Zhao, Department of Biostatistics, Yale School of Public Health New Haven, CT

- Developed a UK Biobank data preprocessing pipeline integrating 1,773 predictors across lifestyle, clinical measurements, environmental exposures, genetics, and imaging, with dimensionality reduction and time-aligned disease outcome construction for downstream multimodal modeling

- Built a real-world acquisition cost framework by mapping measurements to CPT/HCPSC codes and reimbursement schedules
- Contributed to CoFA-MD model training and evaluation, including budget sweeps, baseline comparisons, and ablation analyses of prediction-cost trade-offs

High-Dimensional Mediation of Social Determinants, Brain Connectivity & Youth Mental Health 01/2025 – 08/2026

Supervisor: Prof. Aiyang Zhang, NeuroBioInformatics Lab, School of Data Science, UVA

Charlottesville, VA

- Conducted high-dimensional multivariate mediation analyses in the ABCD cohort to quantify how whole-brain resting-state functional connectivity (rsFC) mediates associations between social determinants of health and youth psychiatric symptom severity
- Reduced multi-domain SDOH variables into interpretable latent components using sparse PCA in R, enabling stable modeling under high-dimensional settings
- Implemented the Directions of Mediation (DM) framework in MATLAB to model whole-brain connectivity as a high-dimensional mediator, with bootstrap-based inference to estimate indirect effects and statistical significance
- Applied longitudinal linear mixed-effects models to examine associations between SDOH latent dimensions and changes in OCD, anxiety, and depressive symptoms over time

Longitudinal Modeling of Mobile Sensing, EMA, and Psychological Well-Being

08/2023 – 05/2025

Supervisor: Prof. Adrienne Wood, Emotion and Behavior Lab, Department of Psychology, UVA

Charlottesville, VA

- Completed a Distinguished Majors Program thesis investigating how primal world beliefs and social exploratory behaviors influence psychological well-being over time, using mobile sensing data.
- Processed 1.2 GB of mobile-sensing data through data cleaning, preprocessing, and visualization using Pandas, NumPy, and Matplotlib in Python.
- Analyzed longitudinal data using linear mixed-effects models implemented in R with the lme4 package, identifying temporal relationships among world beliefs, exploratory behavior, and psychological well-being.

PROFESSIONAL SKILLS

- **Biomedical & Health Informatics:** Electronic health records, Medical imaging data, Wearable and behavioral data, Multimodal data analysis
- **Programming Languages:** Python, R, SQL, MATLAB
- **Language:** English (Proficient), Chinese (Native)